

JavaScript Switch Case Statement - Master Guide

1. Introduction

The switch statement is a control flow statement used to execute one block of code from multiple possible options. It is ideal when comparing a single expression against many fixed values.

2. Syntax

```
switch(expression){  
  case value1:  
    // statements  
    break;  
  case value2:  
    // statements  
    break;  
  default:  
    // statements  
}
```

3. Execution Flow

1. Evaluate the switch expression.
2. Compare each case using strict equality (===).
3. Execute the first matching case.
4. Exit on break, otherwise continue with fall-through.
5. Execute default if no case matches.

4. Supported Expression Types

Type	Supported?	Example
Number	Yes	switch(10)
String	Yes	switch("Admin")
Boolean	Yes	switch(true)
Single-character String	Yes	switch("A")
null	Yes	switch(null)
undefined	Yes	switch(undefined)
Object Reference	Yes	switch(obj)
Symbol	Yes	switch(sym)
BigInt	Yes	switch(10n)

Basic Example

```
let day=2;  
switch(day){  
  case 1: console.log("Monday"); break;  
  case 2: console.log("Tuesday"); break;  
  default: console.log("Invalid");  
}
```

Fall-through

```
switch(1){
  case 1:
    console.log("One");
  case 2:
    console.log("Two");
} // Output: One Two
```

Multiple Cases

```
switch(day){
  case 6:
  case 7:
    console.log("Weekend");
    break;
  default:
    console.log("Weekday");
}
```

String Example

```
switch(role){
  case "Admin":
    console.log("Full Access");
    break;
}
```

Boolean Example

```
switch(isLoggedIn){
  case true:
    console.log("Login successful");
    break;
  case false:
    console.log("Login failed");
}
```

Calculator

```
switch(operator){
  case "+": console.log(a+b); break;
  case "-": console.log(a-b); break;
  case "*": console.log(a*b); break;
  case "/": console.log(a/b); break;
}
```

5. Switch vs If...Else

Feature	switch	if...else
Exact value matching	Excellent	Good
Range comparison	Poor	Excellent
Complex conditions	Poor	Excellent
Readability (many fixed values)	Excellent	Average

6. Common Mistakes

- Expecting switch to use == instead of ===.
- Forgetting break, causing fall-through.
- Using switch for range conditions.
- Using different data types between expression and case values.

7. Best Practices

- Use switch for exact value matching.
- Always include default.
- Use break unless fall-through is intentional.
- Group multiple cases when they share logic.
- Keep each case concise.

8. Interview Questions

1. What comparison operator does switch use?
2. What is fall-through?
3. Is break mandatory?
4. Can default appear anywhere?
5. Can switch use boolean expressions?
6. When should you choose switch over if...else?
7. Can multiple cases execute the same block?
8. What happens if no case matches and there is no default?
9. Can case labels contain expressions?
10. Why doesn't case 10 match switch('10')?

9. Practice Exercises

- Print weekday using switch.
- Build a calculator.
- Display month name from month number.
- Print user permissions based on role.
- Convert numeric grade to letter grade.

10. One-Page Summary

- ✓ switch compares using ===
- ✓ Best for exact value matching
- ✓ break exits only the switch block
- ✓ Missing break causes fall-through
- ✓ default is optional but recommended
- ✓ Use if...else for ranges